

Introduction to Data Visualization with R

Linus Hagemann

ggplot

ggplot is the most common approach for creating data visualizations with R.

Source: Large parts of this section are adapted from the [first chapter of *R for Data Science*](#).

ggplot Syntax

Making a data visualization with ggplot starts with

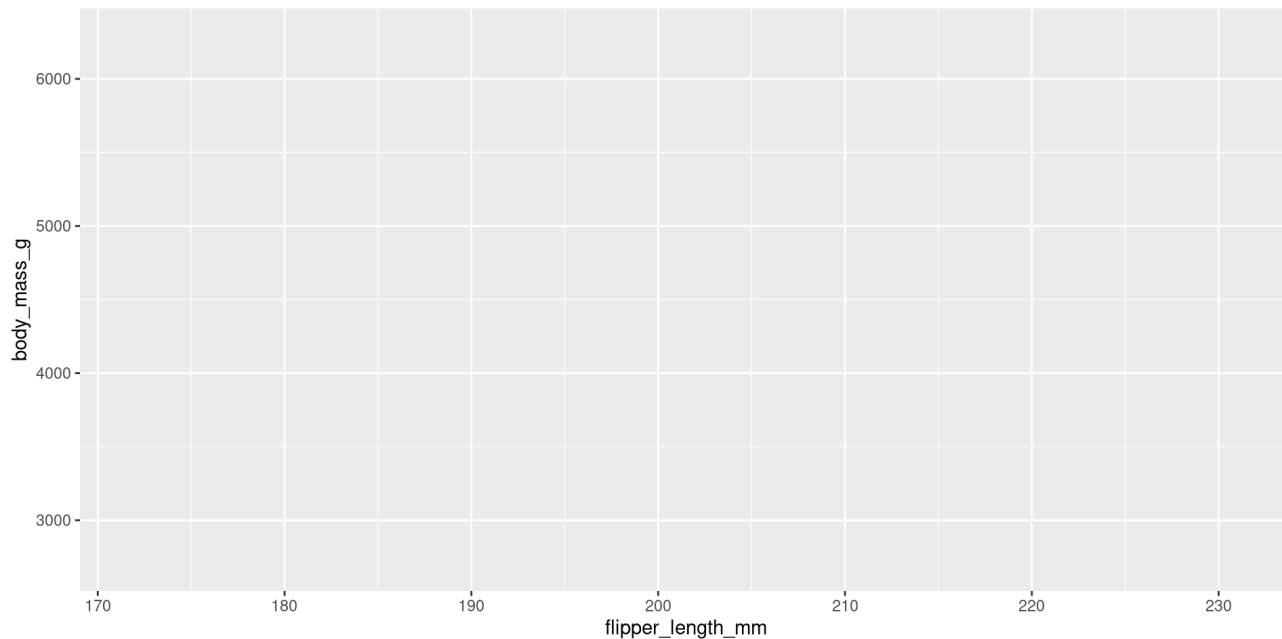
```
1 ggplot(data, aes())
```

where `data` is a dataframe and `aes()` describes how columns in the dataframe will be used in the visualization.

ggplot Aesthetics

For example

```
1 ggplot(penguins, aes(x = flipper_length_mm, y = body_mass_g))
```



will use the x-axis for `flipper_length_mm` and the y-axis for `body_mass_g`.

Aesthetics Options

Common aesthetics include:

- `x`
- `y`
- `color`
- `fill`
- `shape`
- `size`

You can read more about aesthetics in the ggplot documentation [here](#).

Grammar of Graphics

The core idea behind ggplot is that plots are composed by combining components such as data, coordinate systems, and geometries (using `+`). This makes it easy to iterately make data visualizations and keeps the syntax consistent between different kinds of plots.

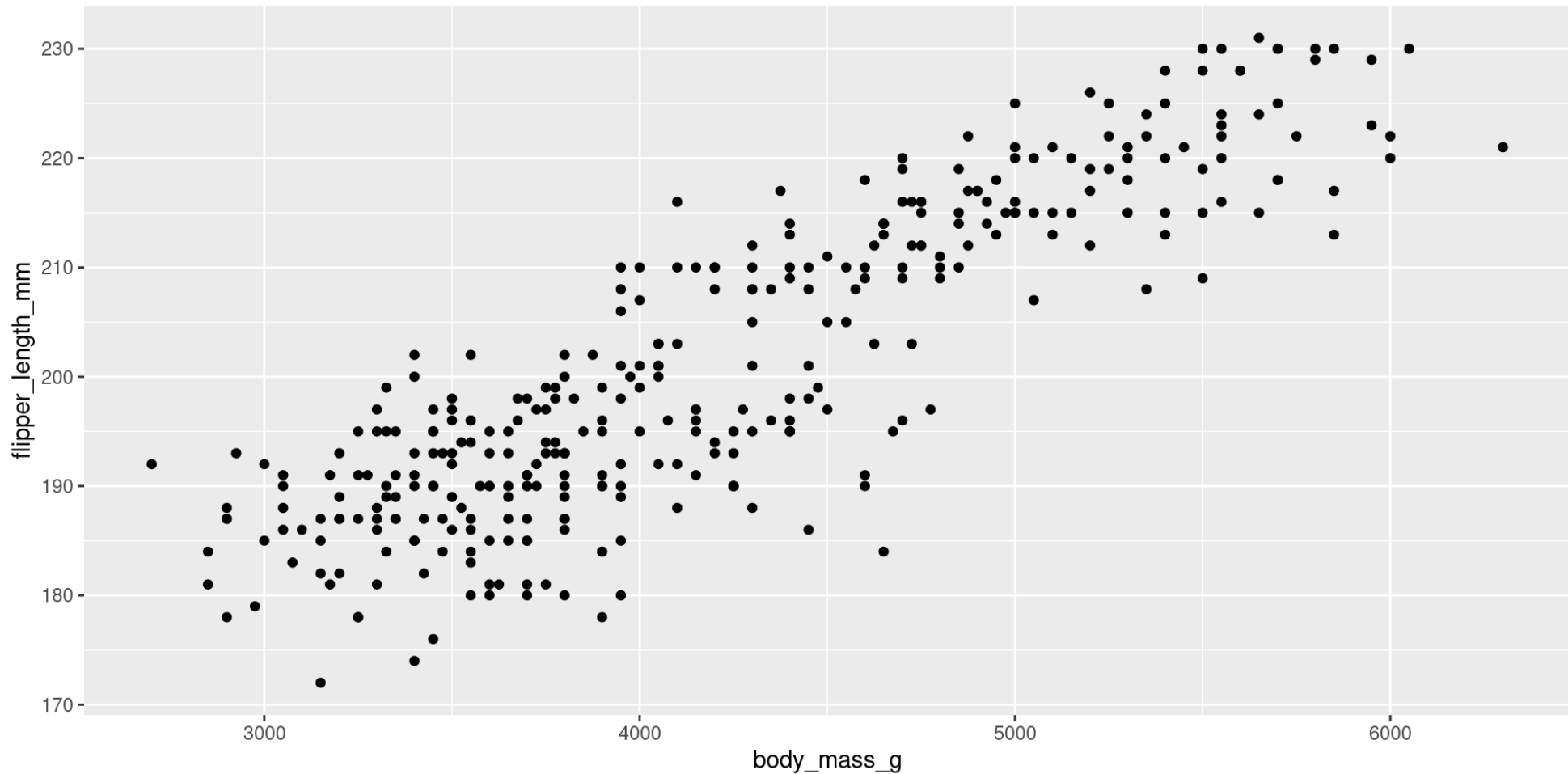
Geometries

The next step in making a plot with ggplot is choosing how to visualize the data. We'll start with a scatter plot.

With ggplot, these choices always start with “geom_”.

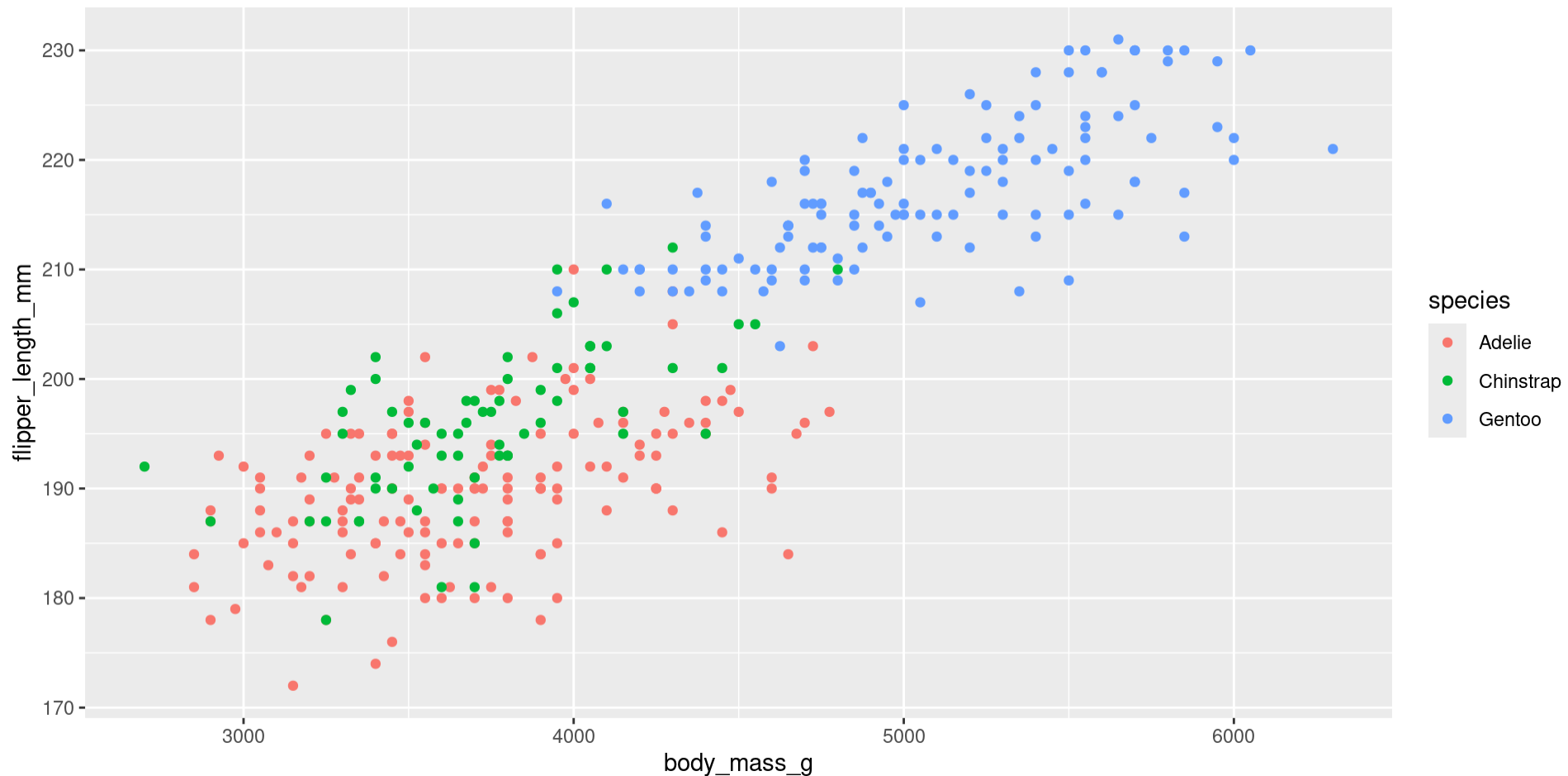
Geometries

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm)) +  
2   geom_point()
```



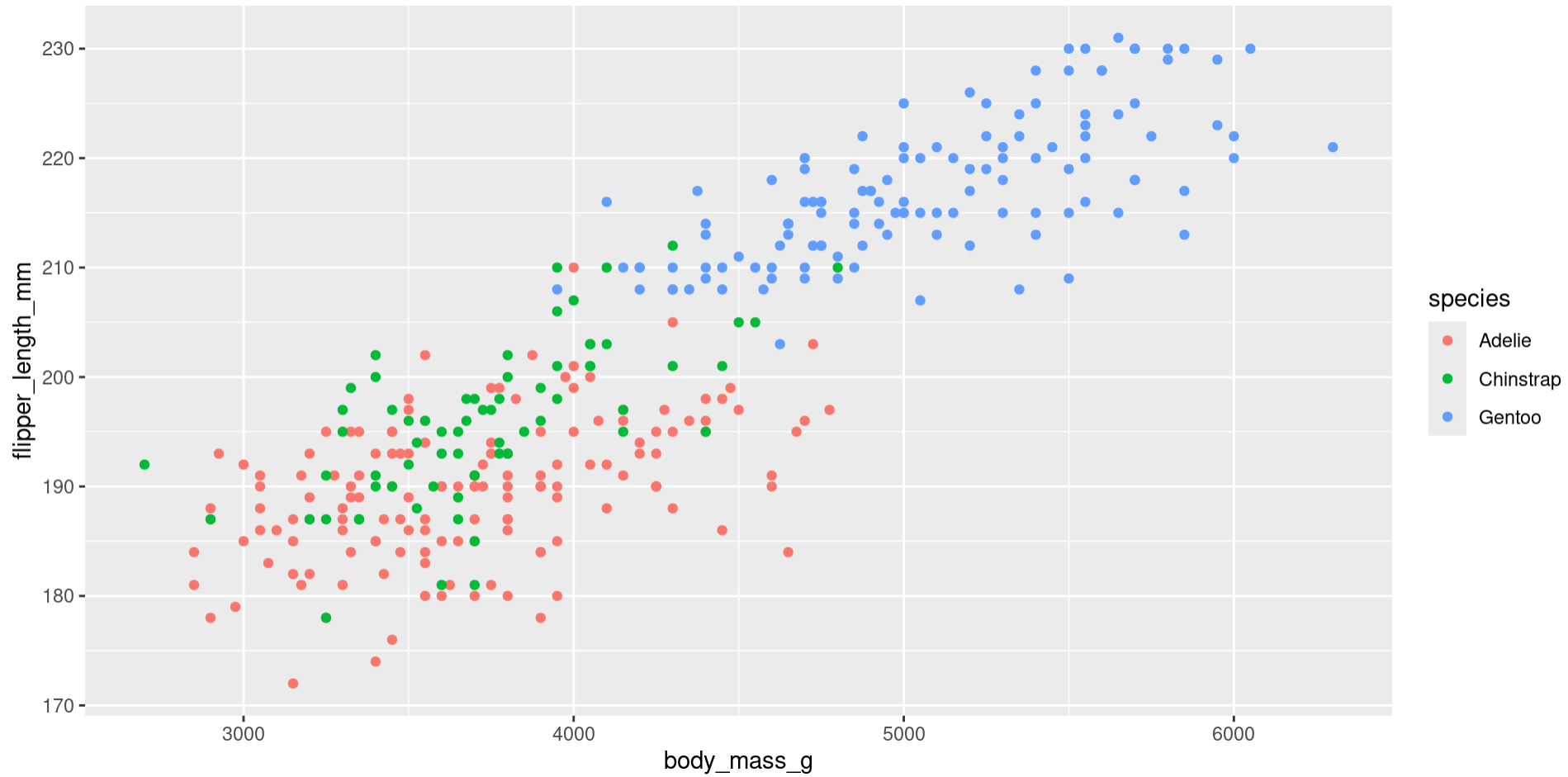
Geometries

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = specie  
2   geom_point())
```



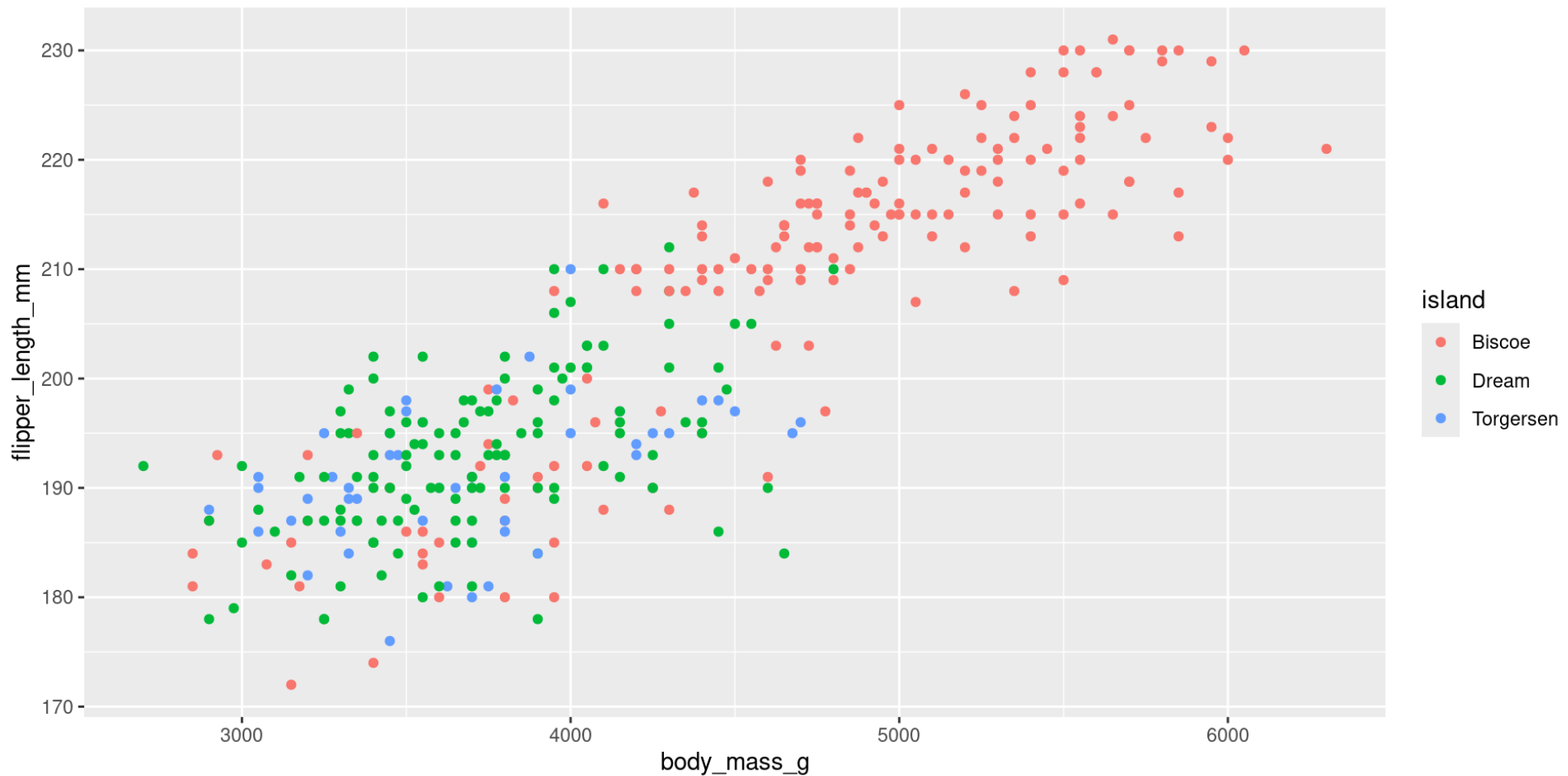
```
1 penguin_plot_default <- ggplot(penguins, aes(x = body_mass_g, y = flipper_l
```

```
2  
3 penguin_plot_default +  
4   geom_point()
```



Geometries

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = island))
2   geom_point()
```



Geometry Options

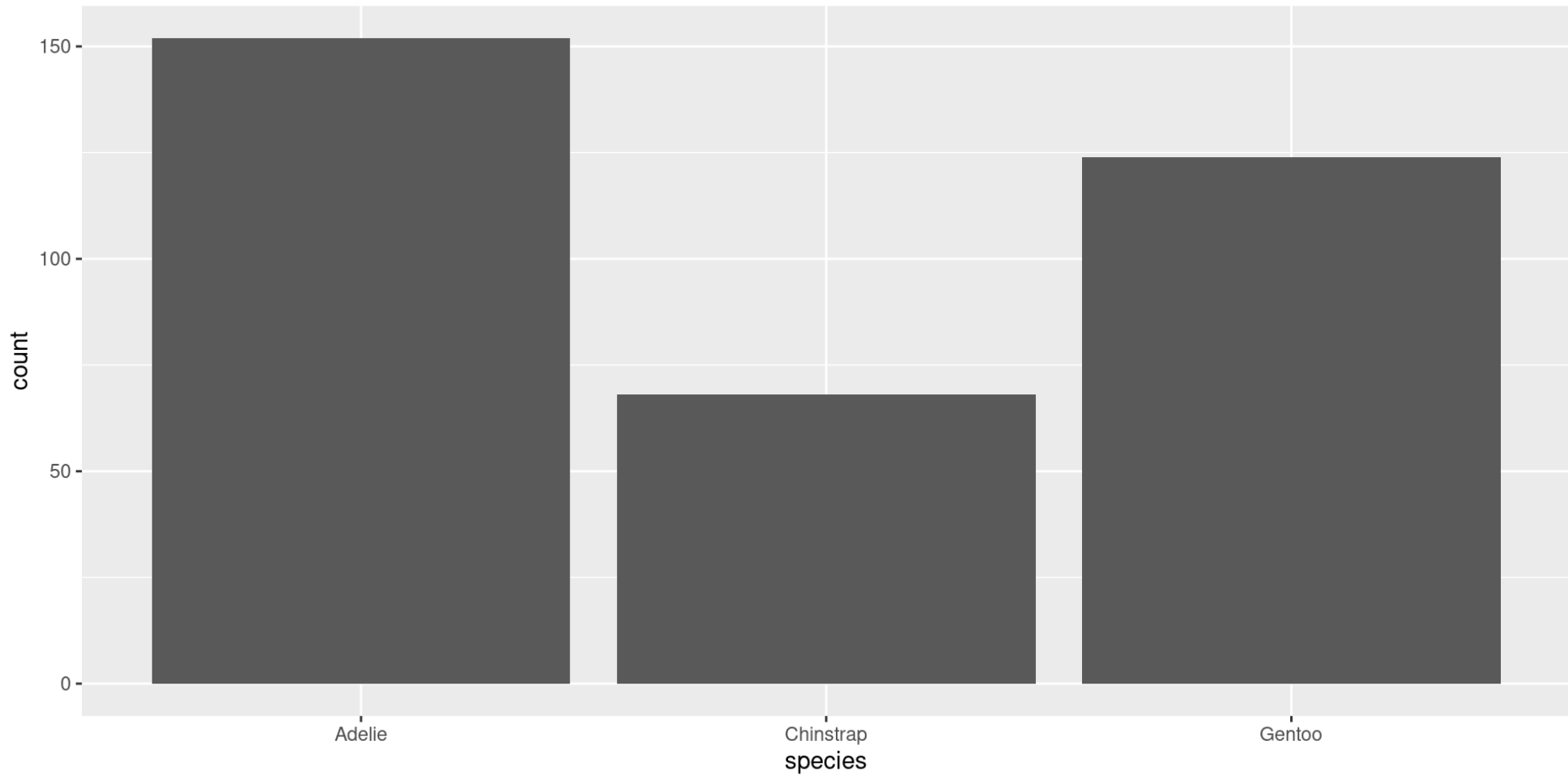
Some of the most common geometries are

- `geom_point()`
- `geom_histogram()`
- `geom_line()`
- `geom_density()`
- `geom_bar()`
- `geom_label()`
- `geom_col()`
- `geom_smooth()`

You can read about all the geometries [here](#).

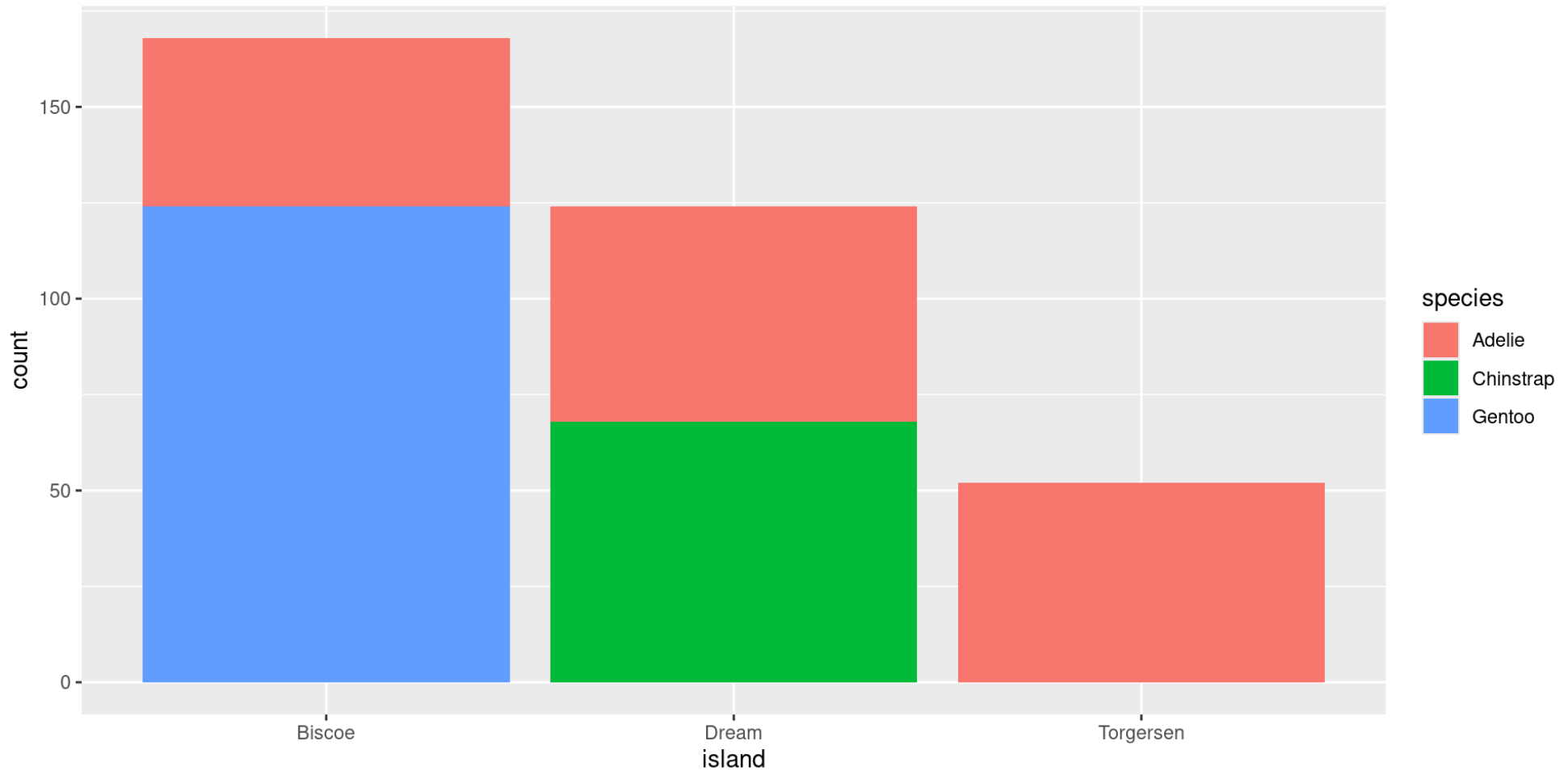
geom_bar() Example

```
1 ggplot(penguins, aes(x = species)) +  
2   geom_bar()
```



Second `geom_bar()` Example

```
1 ggplot(penguins, aes(x = island, fill = species)) +  
2   geom_bar()
```



geom_bar() versus geom_col()

`geom_bar()` only takes an x-axis in `aes()` and uses the number of rows for the y-axis.

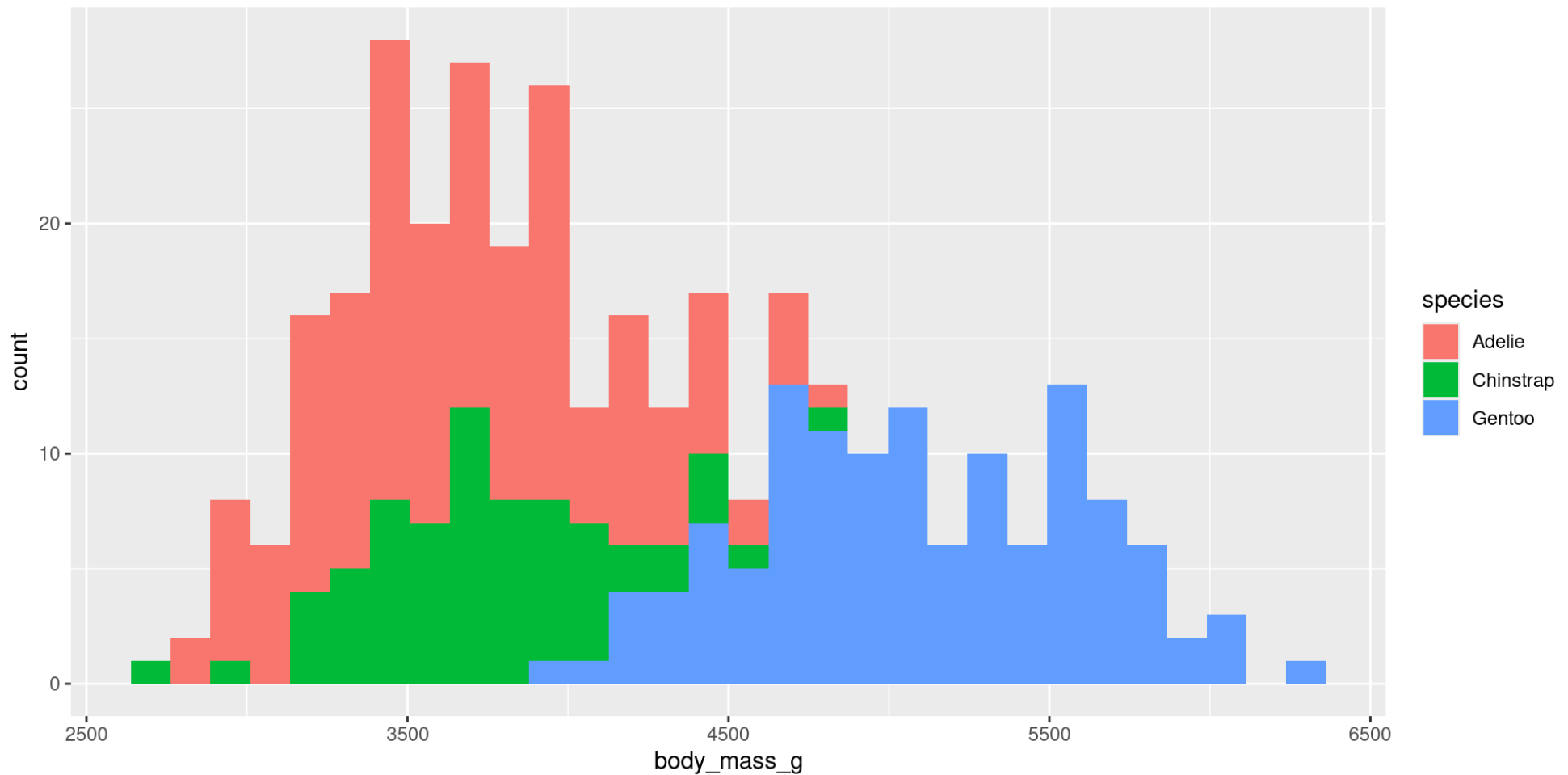
In contrast, `geom_col()` takes a y-axis argument.

geom_histogram()

`geom_histogram()` produces a histogram which visualizes the distribution of a variable/column. Histograms place values into bins (e.g. dividing 0 to 99 into 0-9, 10-19, 20-29... 90-99). The bins are the x-axis and the count of items in each bin is the y-axis.

geom_histogram() Example

```
1 ggplot(penguins, aes(x = body_mass_g, fill = species)) +  
2   geom_histogram()
```

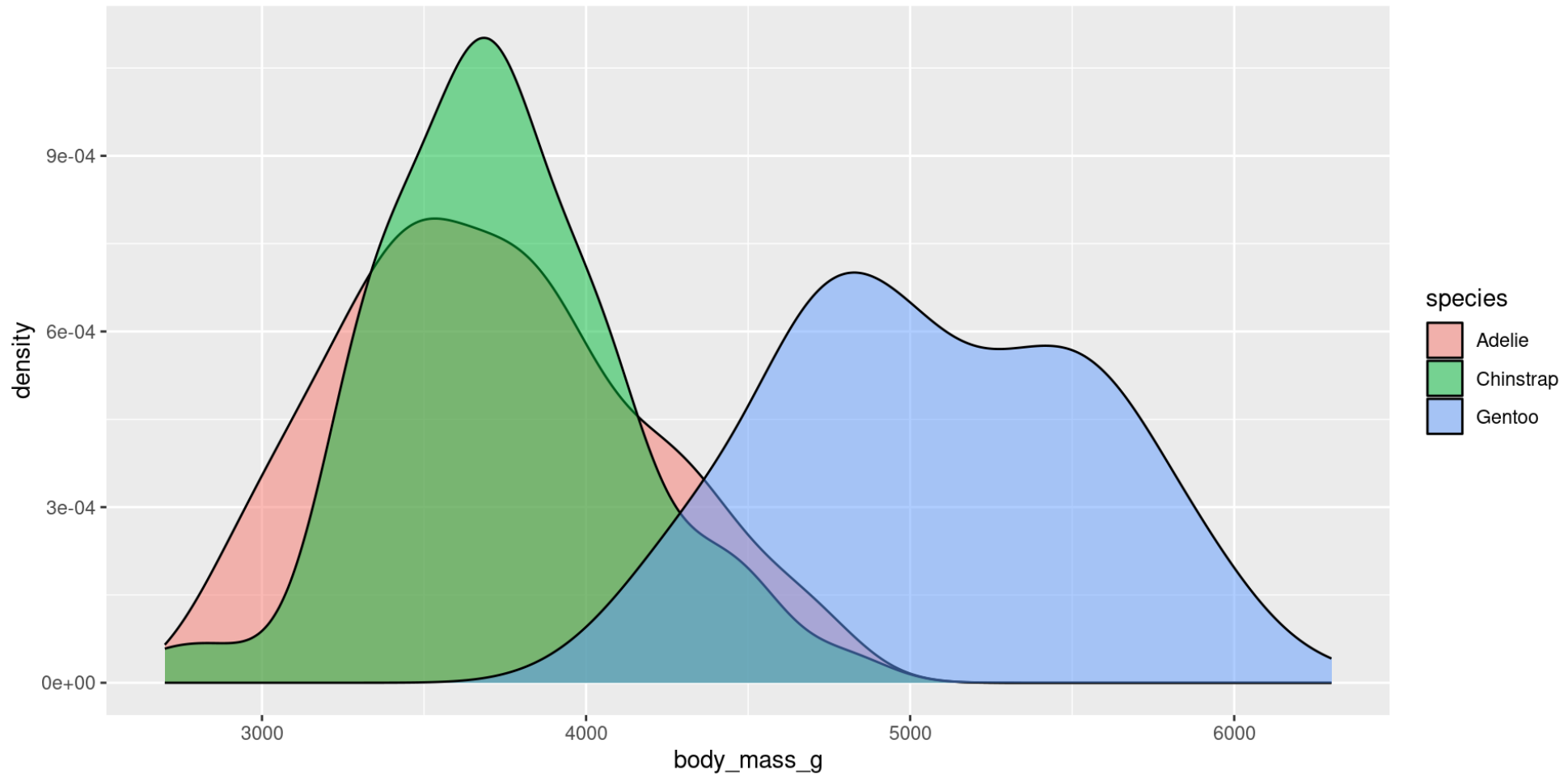


geom_density()

`geom_density()` is another visualization of the distribution of a variable/column. You can think of it as something like a smoothed out histogram.

geom_density() Example

```
1 ggplot(penguins, aes(x = body_mass_g, fill = species)) +  
2   geom_density(alpha = 0.5)
```



geom_smooth()

`geom_smooth()` adds lines of best fit to the plot.

`geom_smooth(method = "lm")` finds the line of best fit with a linear regression (if you don't know what that is, don't worry—it will be covered in Math for Data Science).

geom_smooth() Example

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = specie  
2   geom_point() +  
3   geom_smooth(method = "lm", se = FALSE) # linear regression
```



Labels

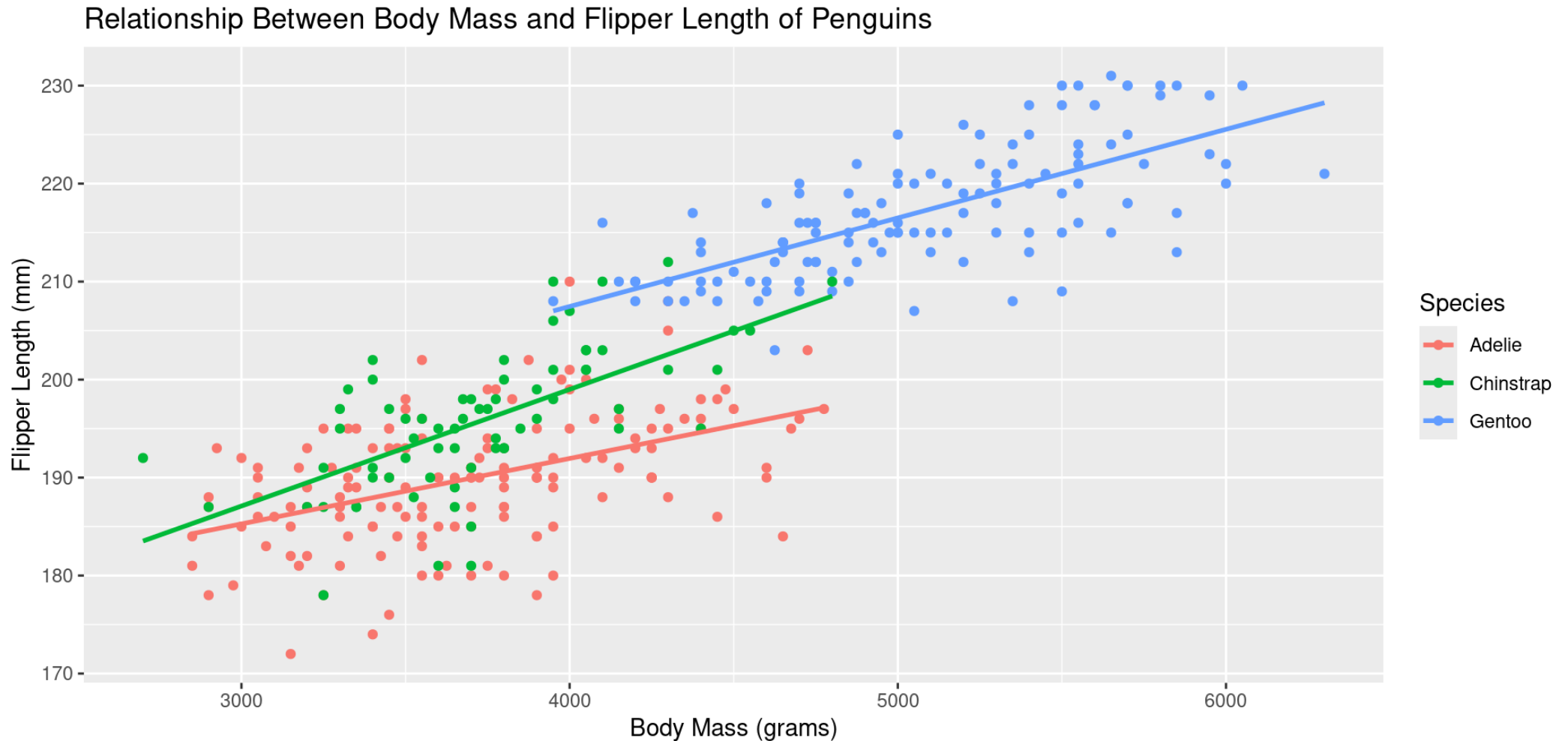
You can label your plot with `labs()` like

```
1 labs(  
2     title = "The plot title",  
3     x = "The x-axis label",  
4     y = "The y-axis label",  
5     caption = "The plot caption"  
6 )
```

Example of Labels

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = specie
2   geom_point() +
3   geom_smooth(method = "lm", se = FALSE) +
4   labs(
5     title = "Relationship Between Body Mass and Flipper Length of Penguins"
6     x = "Body Mass (grams)",
7     y = "Flipper Length (mm)",
8     caption = "Body mass and flipper length are positively correlated for a
9   )
```

Example of Labels



Body mass and flipper length are positively correlated for all three species of penguin

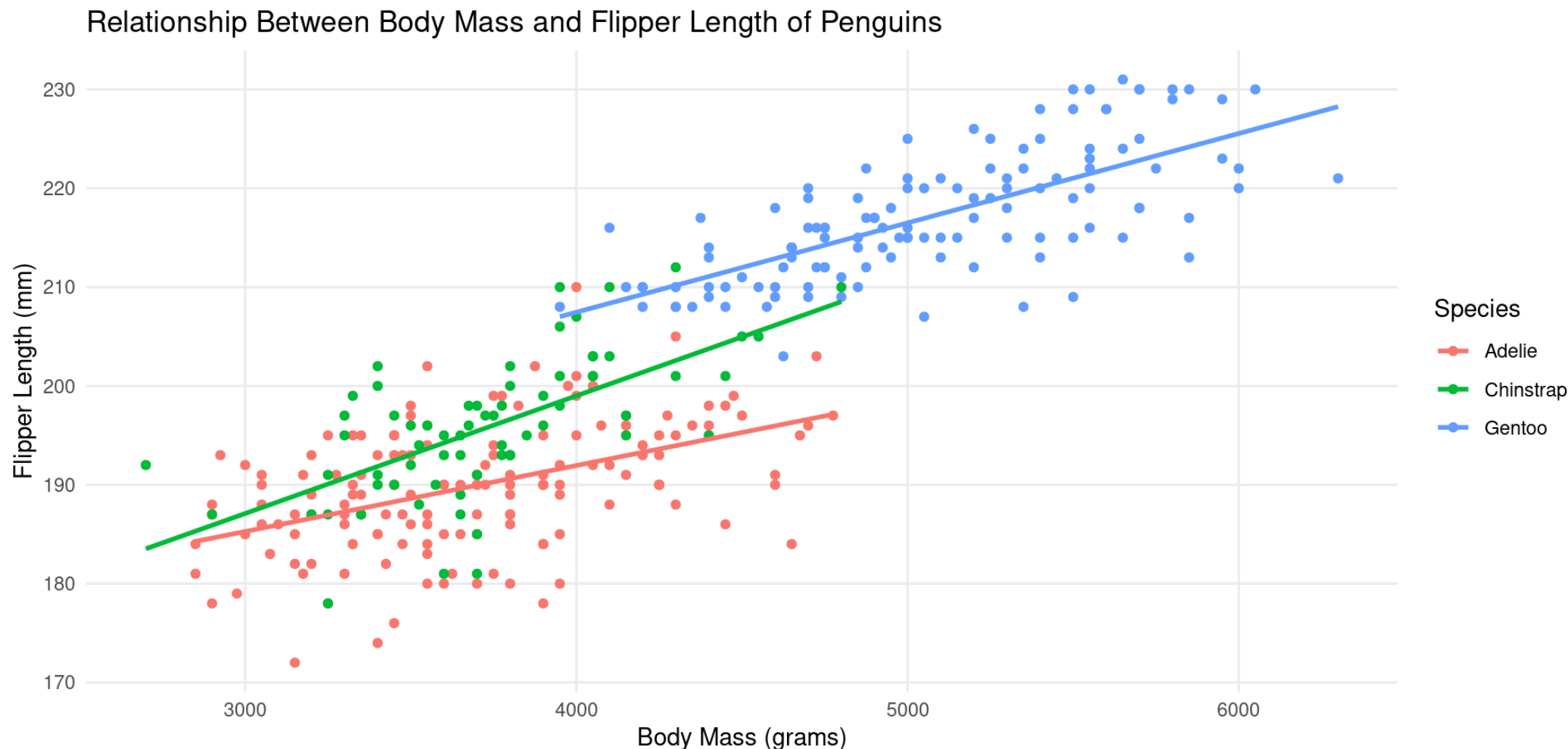
Themes

You can change the appearance of ggplot figures using themes. Complete themes start with `theme_`. You can also adjust the individual theme components with `theme()`.

Theming Example

```
1 ggplot(penguins, aes(x = body_mass_g, y = flipper_length_mm, color = specie
2   geom_point() +
3   geom_smooth(method = "lm", se = FALSE) +
4   labs(
5     title = "Relationship Between Body Mass and Flipper Length of Penguins"
6     x = "Body Mass (grams)",
7     y = "Flipper Length (mm)",
8     color = "Species",
9     caption = "Body mass and flipper length are positively correlated for a
10  ) +
11  theme_minimal() +
12  theme(panel.grid.minor = element_blank())
```

Theming Example



Body mass and flipper length are positively correlated for all three species of penguin

Experiment

Let's take some time and practice making various plots. You can use the penguins dataset or try

- `iris`: Data on the length and width of the sepal and flower of three species of Iris plants.
- `Titanic`: Survival status of passengers on the Titanic by class, sex, and age.
- `trees`: The girth, height, and volume of 31 trees.

You can read about all the datasets built into R with `?datasets`.

Showcase

<https://rustpad.io/#s5yUfk>